

Garbage Classifier Documentation

Project Overview

The **Garbage Classifier** is a Deep Learning-based image classification system that automatically identifies different categories of waste from an uploaded image. The objective of this project is to assist in smart waste management by classifying garbage into its respective categories, making recycling and waste segregation more efficient. The project compares the performance of a custom Convolutional Neural Network (CNN) with Transfer Learning models to achieve higher classification accuracy.

Project Workflow

The overall workflow of the system consists of the following stages:

1. Image Input

The user uploads an image of garbage through the web application. This image acts as the input to the classification pipeline.

2. Image Preprocessing

Before feeding the image into the neural network, several preprocessing operations are performed:

- Image resizing
- Pixel normalization
- Data formatting
- Conversion into tensor format

These preprocessing steps ensure that every image has the same dimensions and distribution before training or prediction.

3. Feature Extraction

The processed image is passed through two pretrained deep learning architectures:

- **ResNet**

- **VGG16**

Both networks extract meaningful visual features from the image.

Instead of relying on a single model, features from both architectures are combined using a **weighted averaging** approach.

This ensemble strategy improves robustness and overall prediction accuracy.

4. Classification

The combined features are passed to the final classification layer, which predicts the garbage category.

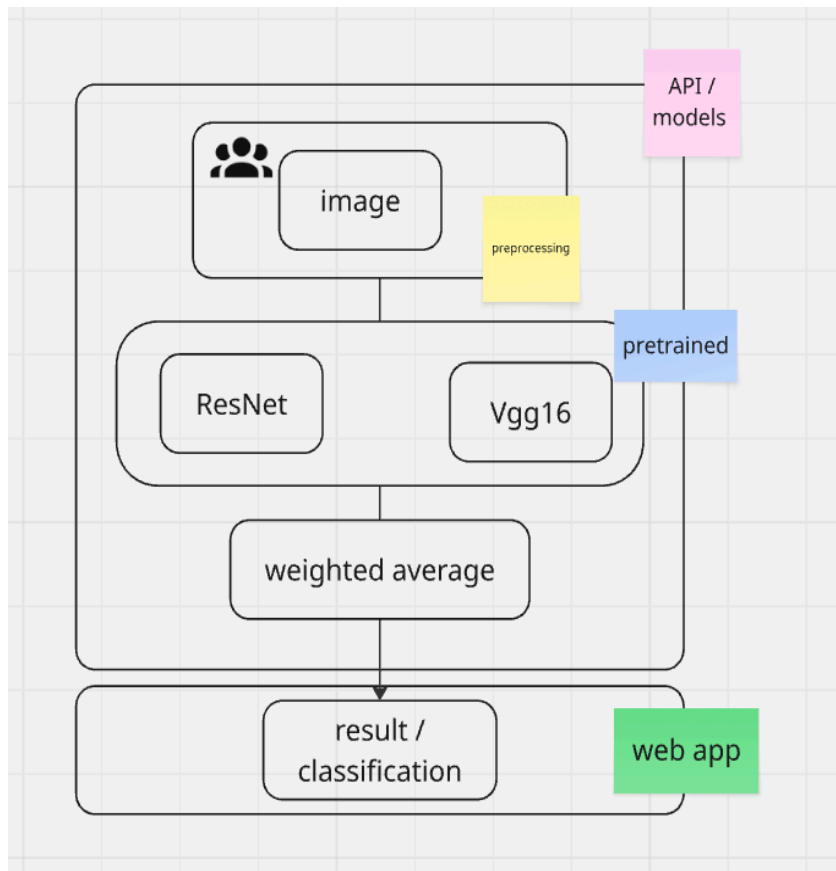
Examples include:

- Plastic
- Paper
- Glass
- Metal
- Cardboard
- Organic Waste

(The exact categories depend on the dataset used.)

5. Result Display

The predicted class is displayed through the web application along with the confidence score. The complete workflow can be summarized as:



This workflow combines preprocessing, transfer learning, and ensemble techniques to improve the classification performance.

Model Parameters

1. Without Transfer Learning (Custom CNN)

A custom CNN was trained from scratch to classify garbage images. Since the model had to learn all features directly from the dataset, it achieved limited performance with an accuracy of **40–50%**.

Parameter	Value
Input Size	224 × 224 × 3
Optimizer	Adam
Learning Rate	0.001
Loss Function	Categorical Crossentropy
Batch Size	32
Activation	ReLU, Softmax
Accuracy	40–50%

2. With Transfer Learning

Transfer Learning was implemented using **ResNet50** and **VGG16** pretrained on ImageNet. The pretrained feature extractors were fine-tuned by training only the last layers, resulting in significantly improved classification performance.

Parameter	Value
Input Size	224 × 224 × 3
Base Models	ResNet50, VGG16
Pretrained Weights	ImageNet
Optimizer	Adam
Learning Rate	0.001
Loss Function	Categorical Crossentropy
Batch Size	32
Activation	ReLU, Softmax
Fine-Tuning	Last layers only
Accuracy	98 and val 92%

Dataset Preparation

The dataset was organized into multiple garbage categories.

The preprocessing stage included:

- Loading images category-wise
- Splitting the dataset into
 - Training Set
 - Validation Set
 - Testing Set
- Applying image preprocessing techniques
- Creating data generators for efficient model training

Proper dataset preparation ensures balanced learning and improves model generalization.

Initial Model (Without Transfer Learning)

Initially, a custom CNN model was trained from scratch.

Observations

- Slow convergence
- Limited feature extraction capability
- Overfitting on training data
- Poor validation accuracy

Performance

The model achieved approximately

40–50% accuracy

which was insufficient for practical use.

Transfer Learning Approach

To improve performance, Transfer Learning was introduced.

Instead of training the entire network from scratch, pretrained ImageNet models were used.

The project utilized:

- ResNet
- VGG16

These pretrained models already contain rich visual features learned from millions of images, allowing the classifier to learn faster and achieve higher accuracy.

The pretrained convolutional layers were used as feature extractors, while new classification layers were added for garbage classification.

97-98 training accuracy + 92 % accuracy in validation

Fine-Tuning Strategy

After loading the pretrained models:

- Earlier convolution layers were frozen.
- Only the final layers were made trainable.
- New dense layers were added.
- The model was trained using the garbage dataset.

Training only the last few layers significantly reduced training time while improving accuracy.

Model Architecture

The final architecture consists of:

- Input Image
- Image Preprocessing
- ResNet Feature Extractor
- VGG16 Feature Extractor
- Weighted Average Layer
- Dense Classification Layer
- Softmax Output Layer

This ensemble approach combines the strengths of multiple pretrained models.

Training Process

During training:

- Training Loss was monitored
- Validation Loss was monitored

- Accuracy curves were observed
- Model weights were updated using backpropagation

Early stopping and validation monitoring help prevent overfitting.

Results

Without Transfer Learning

- Accuracy: **40–50%**
- Poor generalization
- High validation loss

With Transfer Learning

Using pretrained models significantly improved:

- Feature extraction
- Accuracy % **training : 97-98 and validation : 92**
- Model convergence
- Classification accuracy
- Generalization performance

Transfer learning proved to be substantially more effective than training a CNN from scratch.

Advantages of the Proposed System

- Automatic garbage classification
 - Reduced manual effort
 - Better waste segregation
 - Improved recycling efficiency
 - High classification performance using pretrained models
 - Easy deployment as a web application
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Future Improvements

Potential enhancements include:

- Support for more garbage categories
 - Real-time camera-based prediction
 - Mobile application integration
 - Object detection for multiple waste items
 - Cloud deployment
 - Model optimization for edge devices
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Conclusion

The Garbage Classifier demonstrates the effectiveness of Transfer Learning for image classification tasks. While a custom CNN achieved only **40–50% accuracy**, incorporating pretrained architectures such as **ResNet** and **VGG16**, along with a weighted ensemble strategy, and **accuracy 98 % and validation 92 %** significantly enhanced the model's ability to extract meaningful features and produce more reliable predictions. The project highlights how modern deep learning techniques can be applied to smart waste management, enabling automated and efficient garbage classification through a simple web application.